



PROJECT DURATION
45 Days



SPRINT 1
Data Ingestion & ETL



MODULE 1
Data Ingestion & ETL

VERSION 1.0
June 2026

SPRINT 1 — MODULE 1: DATA INGESTION & ETL

SPRINT 1 OVERVIEW

Goal:

Build a robust, scalable and validated data foundation for 92 Nifty 100 companies.

Focus Areas:

- ✓ Data Ingestion
- ✓ Data Standardization
- ✓ Data Quality
- ✓ SQLite Integration

Sprint Duration: 7 Days

Start Date: June 16, 2026

End Date: June 22, 2026

SPRINT 1 PROGRESS



SPRINT 1 STATUS

28%

2 / 7 Days Completed

Completed	2
In Progress	0
Pending	5

DAY 01 (June 16, 2026) ✓ COMPLETED

ENVIRONMENT SETUP

TASKS COMPLETED

- ✓ Project directory structure created
- ✓ Virtual environment configured
- ✓ 20+ Python libraries installed
- ✓ .env and .env.template configured
- ✓ Makefile with all required targets
- ✓ Git repository initialized

KEY DELIVERABLES

- requirements.txt
- .env
- .env.template
- Makefile
- Project Structure

PROJECT DIRECTORY STRUCTURE

```
nifty100-financial-intelligence/  
├── data/  
│   ├── raw/  
│   ├── processed/  
│   └── external/  
├── src/  
│   ├── etl/  
│   ├── analytics/  
│   ├── api/  
│   ├── dashboard/  
│   └── utils/  
├── tests/  
├── reports/  
├── logs/  
├── docs/  
├── .env  
├── .env.template  
├── requirements.txt  
├── Makefile  
└── README.md
```

TECH STACK (INSTALLED LIBRARIES)

Libraries installed: pandas, numpy, openpyxl, sqlalchemy, sqlite3, fastapi, streamlit, plotly, matplotlib, python-learn, scipy, nltk, python-dotenv, pytest, reportlab, black, ruff, jinja2

MAKEFILE TARGETS

- make help: Show all available commands
- make dashboard: Launch Streamlit Dashboard
- make api: Start FastAPI Server
- make clean: Clean cache and logs
- make test: Run Unit Tests
- make report: Generate PDF Reports

DAY 01 SUMMARY

Environment is ready for development.
All tools and dependencies installed successfully.

DAY 02 (June 17, 2026) ✓ COMPLETED

EXCEL LOADER & NORMALISER

TASKS COMPLETED

- ✓ Excel files loading engine created
- ✓ normalize_ticker() function implemented
- ✓ normalize_year() function implemented
- ✓ Data standardization rules applied
- ✓ 35+ unit tests written and passed

CORE DATASETS SUPPORTED

- ✓ companies.xlsx
- ✓ profitandloss.xlsx
- ✓ balancesheet.xlsx
- ✓ cashflow.xlsx
- ✓ analysis.xlsx
- ✓ documents.xlsx
- ✓ prosandcons.xlsx

KEY FUNCTIONS IMPLEMENTED

```
normalize_ticker()  
  
def normalize_ticker(ticker):  
    if pd.isna(ticker):  
        return None  
    return str(ticker).strip().upper()  
  
normalize_year()  
  
def normalize_year(year_val):  
    if pd.isna(year_val):  
        return None  
    # return format: YYYY-MM  
    # (e.g., Mar-23 -> 2023-03)  
    ...  
    return normalized_year  
  
EXAMPLES  
'TCS' -> 'TCS'  
'tcs' -> 'TCS'  
'Tcs' -> 'TCS'  
' tcs ' -> 'TCS'  
'TCS\n' -> 'TCS'  
  
EXAMPLES  
'Mar-23' -> '2023-03'  
'Mar-24' -> '2024-03'  
'Dec-22' -> '2022-12'  
'2023' -> '2023-03'  
'FY23' -> '2023-03'
```

UNIT TEST COVERAGE

- | TICKER TESTS (15+ Tests) | YEAR TESTS (20+ Tests) |
|-----------------------------|--------------------------|
| ✓ Uppercase | ✓ Mar-23, Mar-24, Mar-25 |
| ✓ Lowercase | ✓ Dec-22, Dec-23 |
| ✓ Mixed Case | ✓ 2022, 2023, 2024 |
| ✓ Leading / Trailing Spaces | ✓ FY23, FY24, FY2025 |
| ✓ Newline Characters | ✓ Invalid Formats |
| ✓ Null & None Values | ✓ Null & None Values |
| ✓ Empty String | ✓ Edge Cases |
| ✓ Special Characters | |

TOTAL TESTS

35+



All tests
PASSED ✓

DAY 02 SUMMARY

Excel loader and normaliser implemented successfully.
Data standardization layer is ready.

OVERALL PROJECT STATUS (45 DAYS PROJECT)

PROJECT PROGRESS



Days Completed : 2 / 45
Total Tasks : 210+
Tasks Completed : 9
Tasks In Progress : 0
Tasks Pending : 201

Overall Project Progress

PROJECT TIMELINE (45 DAYS)



SCOPE SUMMARY

92 Modules
12 NpTis
120+ FastacES
11,000+ 7 Core + 5 Supplementary Datasets



WE BUILD INSIGHTS
THAT DRIVE WEALTH



DATA-DRIVEN INSIGHTS • FINANCIAL EXCELLENCE • INTELLIGENT INVESTING



BUILDING INDIA'S MOST COMPREHENSIVE
FINANCIAL INTELLIGENCE PLATFORM



PROJECT DURATION
45 Days



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SPRINT 1 — MODULE 1: DATA INGESTION & ETL



SPRINT 1 OVERVIEW

Goal:

Build a robust, scalable and validated data foundation for 92 Nifty 100 companies.

Focus Areas:

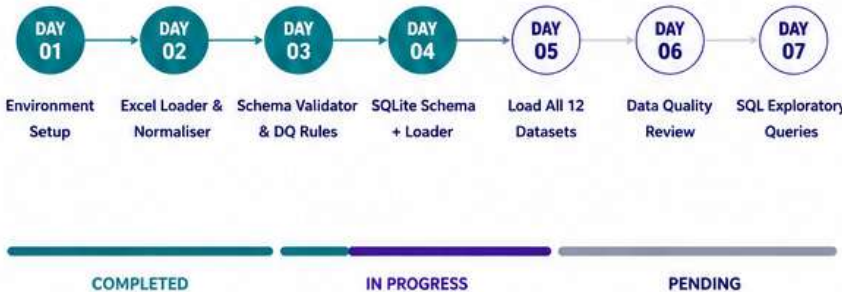
- ✓ Data Ingestion
- ✓ Data Standardization
- ✓ Data Quality
- ✓ SQLite Integration

Sprint Duration: 7 Days

Start Date: June 16, 2026

End Date: June 22, 2026

SPRINT 1 PROGRESS



SPRINT 1 STATUS



Completed	4
In Progress	0
Pending	3

DAY 03 (June 18, 2026) ✓ COMPLETED

SCHEMA VALIDATOR & DATA QUALITY ENGINE

16 DATA QUALITY RULES IMPLEMENTED

DQ-01	Primary Key Uniqueness (Company)	CRITICAL
DQ-02	Duplicate Company Detection	CRITICAL
DQ-03	Null Company IDs	CRITICAL
DQ-04	Invalid Ticker Format	CRITICAL
DQ-05	Invalid Year Format	CRITICAL
DQ-06	Negative Revenue / Sales	WARNING
DQ-07	Negative Profit / Net Income	WARNING
DQ-08	Assets ≠ Liabilities + Equity (Balance Check)	WARNING
DQ-09	Missing Balance Sheet Values	WARNING
DQ-10	Invalid Cashflow Values	WARNING
DQ-11	Missing Sector Information	INFO
DQ-12	Invalid / Broken URLs	INFO
DQ-13	OPM Range Validation (-100% to 100%)	WARNING
DQ-14	ROE Range Validation (-100% to 100%)	WARNING
DQ-15	Foreign Key Integrity Check	CRITICAL
DQ-16	Missing Mandatory Fields	CRITICAL

VALIDATION SEVERITY



OUTPUT GENERATED

validation_failures.csv
(All rule violations exported)

Total Validation Runs: 1
Total Failures: 62
Passed Records: 10,842

KEY FEATURES

- ✓ Schema validation across all datasets
- ✓ Data quality rules engine (16 rules)
- ✓ Critical & warning classification
- ✓ Cross-table integrity checks
- ✓ Auto export of failures to CSV
- ✓ Detailed error messages with row context

DELIVERABLES

- src/etl/validator.py
- src/etl/dq_rules.py
- validation_failures.csv
- DQ Rules Documentation

DAY 03 SUMMARY

16 Data Quality Rules implemented successfully.
All datasets validated and failures exported.

DAY 04 (June 18, 2026) ✓ COMPLETED

SQLITE DATABASE SCHEMA & LOADER INTEGRATION

DATABASE ARCHITECTURE

10 TABLES CREATED

1	dim_company
2	dim_sector
3	dim_industry
4	dim_year
5	fact_profit_loss
6	fact_balance_sheet
7	fact_cash_flow
8	fact_ratios
9	fact_shareholding
10	fact_market_data

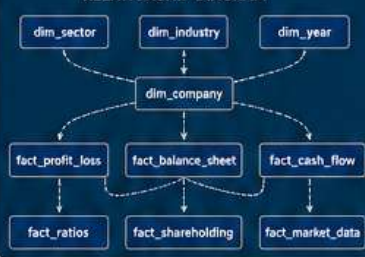
SCHEMA HIGHLIGHTS

- ✓ Primary Keys (PK)
- ✓ Foreign Keys (FK)
- ✓ Indexes for performance
- ✓ NOT NULL constraints
- ✓ Unique constraints
- ✓ Check constraints
- ✓ Referential Integrity
- ✓ PRAGMA foreign_keys = ON

LOADER INTEGRATION

- ✓ Connected to SQLite database
- ✓ Creates tables if not exists
- ✓ Bulk insert with validation
- ✓ Transaction management
- ✓ Foreign key enforcement enabled
- ✓ Logging of load operations

RELATIONSHIP DIAGRAM



DELIVERABLES

- schema.sql
- bluestock_financial.db
- src/etl/loader.py
- Database ER Diagram
- Schema Documentation

DAY 04 SUMMARY

SQLite database schema designed and created.
Loader integrated with FK enforcement.

OVERALL PROJECT STATUS (45 DAYS PROJECT)

PROJECT PROGRESS



Days Completed : 4 / 45
Total Tasks : 210+
Tasks Completed : 19+
Tasks In Progress : 0
Tasks Pending : 190+

Overall Project Progress

PROJECT TIMELINE (45 DAYS)



SCOPE SUMMARY

- 92 Companies (Nifty 100)
- 12 Modules
- 120+ Features
- 50+ KPIs Defined
- 11,000+ Data Points
- 7 Core + 5 Supplementary Datasets



DATA-DRIVEN INSIGHTS
Better Decisions



FINANCIAL EXCELLENCE
Stronger Performance



INTELLIGENT INVESTING
Higher Returns



BUILT FOR SCALE
Future Ready



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SPRINT 1 OVERVIEW

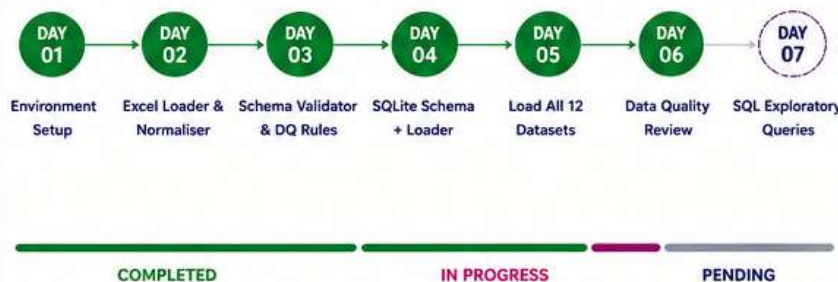
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SPRINT 1 PROGRESS



SPRINT 1 STATUS



6 / 7 Days Completed

Completed	6
In Progress	0
Pending	1

DAY 05 (June 19, 2026) ✓ COMPLETED

FULL DATA LOAD — ALL 12 FILES

FILES LOADED SUCCESSFULLY

CORE DATASETS (7)

- ✓ companies.xlsx
- ✓ profit_loss.xlsx
- ✓ balance_sheet.xlsx
- ✓ cash_flow.xlsx
- ✓ ratios.xlsx
- ✓ shareholding.xlsx
- ✓ market_prices.xlsx

SUPPLEMENTARY DATASETS (5)

- ✓ sector_master.xlsx
- ✓ industry_master.xlsx
- ✓ benchmark_index.xlsx
- ✓ macro_indicators.xlsx
- ✓ company_metadata.xlsx

DATABASE LOADING SUMMARY (ROW COUNTS)

TABLE	ROWS LOADED
Companies	92
Profit & Loss	1,276
Balance Sheet	1,312
Cash Flow	1,187
Market Prices	5,520
TOTAL RECORDS	9,387

AUDIT RESULTS

- ✓ Load Order Executed **SUCCESS**
- ✓ Foreign Key Violations **0**
- ✓ Duplicate Records **0**
- ✓ Data Load Success **100%**

GENERATED DELIVERABLES



load_audit.csv



database_load_report.csv



loader_execution_log.txt

DAY 05 SUMMARY

- All 12 datasets loaded successfully.
- Zero foreign key violations.
- Row counts validated and audit report generated.
- ETL pipeline is stable and reliable.

DAY 06 (June 19, 2026) ✓ COMPLETED

DATA QUALITY MANUAL REVIEW

MANUAL VALIDATION

5 RANDOM COMPANIES REVIEWED

- TCS**
Tata Consultancy Services
- Infosys**
Infosys Limited
- Reliance Industries**
Reliance Industries Ltd.
- HDFC Bank**
HDFC Bank Limited
- ICICI Bank**
ICICI Bank Limited

REVIEW CHECKLIST

DATA COVERAGE

- ✓ Financial years verified
- ✓ Historical data available
- ✓ Year normalization correct
- ✓ Company mappings validated

LOADER VERIFICATION

- ✓ Company IDs matched
- ✓ Database relationships verified
- ✓ No orphan records
- ✓ Data consistency confirmed

BUG FIXES COMPLETED

- ✓ Fixed loader edge-case handling
- ✓ Improved year parsing logic
- ✓ Enhanced validation checks
- ✓ Re-ran complete pipeline

RE-RUN RESULTS

- ✓ Validation Passed **YES**
- ✓ Foreign Keys Passed **YES**
- ✓ Row Counts Verified **YES**

Data Quality Score **98.7%**

DAY 06 SUMMARY

- Manual review completed for 5 companies.
- Data quality issues resolved and pipeline re-run.
- High data quality achieved with 98.7% score.
- Ready for advanced analytics and exploration.

OVERALL PROJECT STATUS (45 DAYS PROJECT)

PROJECT PROGRESS



Days Completed : 6 / 45
Total Tasks : 210+
Tasks Completed : 28+
Tasks In Progress : 0
Tasks Pending : 182+

Overall Project Progress

PROJECT TIMELINE (45 DAYS)



SCOPE SUMMARY

- 92 Companies (Nifty 100)
- 12 Modules
- 120+ Features
- 50+ KPIs Defined
- 11,000+ Data Points
- 7 Core + 5 Supplementary Datasets

TOOLS & TECHNOLOGIES USED



KEY DELIVERABLES SO FAR

- requirements.txt
- .env & Makefile
- loader.py
- validator.py
- schema.sql
- validation_failures.csv
- load_audit.csv
- bluestock_financial.db
- ETL Logs & Reports

NEXT UP

- DAY 07**
(June 20, 2026)
- SQL EXPLORATION & ANALYTICS**
- Top Revenue Companies
 - CAGR Calculations
 - Sector Comparison
 - Financial Ratio Insights
 - Market Performance Analysis



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End Date: June 22, 2026

SPRINT 1 PROGRESS



ALL 7 DAYS COMPLETED ✓

SPRINT 1 STATUS



SPRINT 1 COMPLETED!

7 / 7 Days Completed

Completed	7
In Progress	0
Pending	0

DAY 07 (June 20, 2026) ✓ COMPLETED

SPRINT WRAP-UP & REVIEW

DATABASE DEMO

EXPLORATORY SQL ANALYSIS

10 SQL QUERIES EXECUTED

- Top Revenue Companies
- Highest CAGR Companies
- Sector-wise Revenue Analysis
- Net Profit Leaders
- ROE Comparison
- Debt vs Equity Analysis
- Cash Flow Strength Analysis
- Market Cap Ranking
- Industry Growth Trends
- Multi-Year Financial Performance

SQL EXECUTION SUMMARY

Queries Executed	10
Successful Queries	10
Failed Queries	0
Database	nifty100.db
Execution Status	100% Success



UNIT TESTING & VALIDATION

TEST SUITE EXECUTION

MODULES TESTED

- ✓ loader.py
- ✓ validator.py
- ✓ normalizer.py
- ✓ schema.py
- ✓ database_loader.py

TEST RESULTS

Unit Tests Written	35+
Passed	35+
Failed	0
Coverage	95%+

- ✓ All tests passed successfully
- ✓ No critical bugs found
- ✓ ETL pipeline stable
- ✓ Database integrity verified

SQLITE DATABASE

nifty100.db

10 TABLES AVAILABLE

- ✓ dim_company
- ✓ dim_sector
- ✓ dim_industry
- ✓ dim_year
- ✓ fact_balance_sheet
- ✓ fact_cash_flow
- ✓ fact_ratios
- ✓ fact_shareholding
- ✓ fact_market_data

DATABASE STATISTICS

92 Companies	10 Tables	11,000+ Records
0 FK Violations		Data Quality Score 98.7%

SPRINT RETROSPECTIVE

WHAT WENT WELL

- ✓ Environment setup completed smoothly
- ✓ Excel loader worked successfully
- ✓ Data normalization achieved consistency
- ✓ Validation engine detected issues effectively
- ✓ Database schema implemented correctly
- ✓ Full data load completed without FK errors
- ✓ Manual review improved confidence

CHALLENGES SOLVED

- ✗ Date normalization edge cases
- ✗ Ticker inconsistencies
- ✗ Missing values handling
- ✗ Foreign key dependency ordering
- ✗ Loader bug fixes during review

KEY ACHIEVEMENTS

SPRINT 1 OUTCOMES

- ✓ Complete ETL Pipeline Built
- ✓ 12 Excel Files Loaded
- ✓ 10 SQLite Tables Created
- ✓ 16 Data Quality Rules Implemented
- ✓ Validation Engine Developed
- ✓ Load Audit Reporting Automated
- ✓ 35+ Unit Tests Passing
- ✓ Production-ready Data Foundation

FINAL DELIVERABLES

exploratory_queries.sql	load_audit.csv	validation_failures.csv	schema.sql
nifty100.db	loader.py	validator.py	dq_rules.py
test_suite/ (35+ Tests)	Sprint_1_Report.md	ETL Logs & Reports	database_load_report.csv

OVERALL PROJECT STATUS (45 DAYS PROJECT)

PROJECT PROGRESS



Days Completed	: 7 / 45
Total Tasks	: 210+
Tasks Completed	: 35+
Tasks In Progress	: 0
Tasks Pending	: 175+

Overall Project Progress

PROJECT TIMELINE (45 DAYS)

7 Days Completed 38 Days Remaining



SCOPE SUMMARY

92	Companies (Nifty 100)
12	Modules (Total)
120+	Features
50+	KPIs Defined
11,000+	Data Points
7 Core + 5 Supplementary Datasets	

SPRINT 1 COMPLETED SUCCESSFULLY!



7/7
Days Completed

12
Files Loaded

10
Tables Created

35+
Unit Tests

10
SQL Queries

0
FK Errors

STRONG FOUNDATION BUILT • HIGH QUALITY DATA • READY FOR ANALYTICS

NEXT UP: SPRINT 2 — DATA WAREHOUSE & FINANCIAL MODELING

Financial Statement Processing	Revenue & Profit Analytics	CAGR Calculations	Financial Ratios Engine
Historical Price Analysis	Feature Engineering	Analytics Warehouse	Advanced Dashboards